When Does Structure Help a Magnitude-Only Channel Estimator? Aperture, Effective Rank, and a Prospective Test

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Abstract—**A receiver that measures only field magnitude, with a known additive reference, makes multi-user channel estimation a biased phase-retrieval problem whose Fisher information is rank one per measurement and block diagonal across receive elements. Normalised per-element information therefore does not grow with array size, and exploiting aperture requires cross-element structure. On a uniform linear array the Hankel lifting of each channel column supplies exactly such structure, with a capacity of $\lceil N/2 \rceil$ ranks. This predicts, before any measurement, that a rank prior helps only when the channel occupies a small fraction of that capacity — and therefore that it must *hurt* at small aperture. Both follow: enforcing the structure inside an exact expectation–maximization Gerchberg–Saxton iteration, with the order chosen from a held-out pilot residual, gains $+2.93$ dB at $N = 32$ and loses 0.25 dB at $N = 8$ in summed squared error — means over the same twelve operating points at each aperture — while the unstructured baseline moves by at most 0.163 dB across the same range. The reversal is visible only in that statistic: under a paired per-trial median the same $N = 8$ cells give $+0.225$ dB, because the order selector abstains in 42% of trials and those trials are bit-identical to the baseline. Both are reported throughout. The governing variable is not path count but effective rank relative to capacity; we use that relation to predict the gain on two channel distributions it was not fitted to, to errors of 0.13 and 0.315 dB. Simulation only; the bounds do not govern biased estimators.**

Index Terms—**Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér–Rao bound, effective rank.**

I. Introduction

ATOMIC receivers based on Rydberg states read out radio fields through an optical transition whose response is proportional to field *magnitude*, discarding phase [1], [2]. Multi-user channel estimation is therefore biased phase retrieval, and Gerchberg–Saxton-type alternating projection [3] is the natural solver; recent work unrolls it into a trained network [4], and a rank projection has been placed inside projected gradient descent for the same problem [5].

The atomic receiver enters this paper only through the observation model — a magnitude-only measurement with a known additive reference, (1). Nothing below depends on the vapour-cell physics, so the results apply to any array receiver with that measurement, of which the Rydberg receiver is the motivating instance.

This paper is organised around a single argument, stated first and tested afterwards.

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Step 1: unstructured solvers cannot use aperture. Under the unconstrained row-wise parameterisation of (1), row n of $\mathbf{GS}$ enters only measurement row n, so the Fisher information is block diagonal across receive elements (§III). The per-element estimation problem is unchanged by adding elements, and normalised per-element information does not grow with N.

Step 2: aperture is therefore usable only through cross-element structure. On a ULA each path contributes one spatial complex exponential, so the Hankel lifting of every channel column is rank deficient [10], [11]. That is one such structure, not the only one.

Step 3: the structure has a finite capacity. The lifting admits at most $r_{\max}(N) = \lceil N/2 \rceil$ ranks (§II). A rank prior is informative only when the channel occupies a small fraction of it.

Step 4: so the prior must fail, and then harm, at small aperture. At $N = 8$ the capacity is 4 while the simulated path counts are $L_k \sim \mathcal{U}\{3, \ldots, 7\}$; the constraint is frequently vacuous and, when active, truncates genuine components. *A sign reversal at small N is predicted by the argument, not discovered afterwards*, and §IV-B reports it as such — in summed squared error, the quantity a bound governs. It does not appear in a median, for a reason §IV-D makes precise.

The remaining difficulty is knowing *when* "a small fraction of capacity" holds. A qualitative appeal to sparsity is unusable: a 16-path channel on a 32-element array is algebraically full rank yet has effective rank 8.73, and its lifting is compressible with nothing sparse about it. §IV-D replaces path count with effective rank relative to capacity, and §IV-E tests that replacement prospectively.

Contributions.

1) The information argument above, with a Rician bound in which each magnitude measurement contributes *rank-one* information and a geometry-constrained bound in tangent-space form.
2) The aperture result, as a test of a prediction made in advance.
3) Replacement of path count by $r_{\text{eff}}/r_{\max}$, with $r_{\max} = \lceil N/2 \rceil$ derived, separating what the *prior* does from what the *estimator* does.
4) Two cross-generator predictions, registered before measurement.

II. System Model and Estimator

An N-element ULA serves K single-antenna users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \qquad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2), \tag{1}$$

the modulus applied after the noise, so (1) cannot be linearised. Each user's channel is a sum of L_k plane waves on the array manifold,

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \qquad \psi_{\ell,k} = \pi \sin\theta_{\ell,k}, \tag{2}$$

with $L_k \sim \mathcal{U}\{3, 7\}$ [5] and $\theta \sim \mathcal{U}(-90°, 90°)$. The reference-to-signal ratio (RSR) is the reference power over the *single-user* received power.

Structure and capacity. Substituting (2) into the $(N-p)\times(p+1)$ Hankel lifting $\mathcal{H}_p$ separates the exponential in the two indices, so each path contributes one rank-one outer product and $\operatorname{rank}\mathcal{H}_p(\mathbf{g}_k) \le L_k$. Since $\mathcal{H}_p$ is $(N-p)\times(p+1)$,

$$r_{\max}(N) = \max_p \min(N-p,\, p+1) = \lceil N/2 \rceil, \tag{3}$$

and if $L_k \ge r_{\max}(N)$ the constraint is vacuous. K **cancels:** the structural model uses $3\sum_k L_k$ parameters against $2NK$ unstructured, and the ratio is independent of K, so the gain should be K-invariant at fixed pilot adequacy — tested in §IV-F.

The estimator. HS-GS applies, after every exact EM-GS update, a Cadzow projection $\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^\dagger(\Pi_r(\mathcal{H}_p(\mathbf{g})))$, where Π_r truncates to r leading singular values and $\mathcal{H}_p^\dagger$ averages antidiagonals. One application is *not* idempotent — it is one step of an alternating projection between the rank set and the Hankel set — so the sweep count is part of the operator's definition. **The operator settings are not uniform across §IV**: every result below is run at one of the four configurations of Table I, and each number is tagged with the one that produced it. All four use $n_{\mathrm{cz}} = 4$; the iteration count T and the selection budget differ. With the projection disabled HS-GS reproduces EM-GS to $\max|\mathrm{diff}| = 0$, so the projection is provably the only difference.

Order selection uses no oracle. A fraction $\nu = 0.3$ of the pilot columns is held out; each candidate $r \in \{1, \ldots, r_{\max}\}$ is scored by the held-out magnitude residual and the minimiser is used. The in-sample residual cannot serve: constraining $\mathbf{G}$ can only worsen the fit to the pilots used for fitting, so an in-sample criterion always prefers the largest candidate. When the selector returns $\hat{r} \ge r_{\max}$ the projection is the identity and the estimator *abstains*, falling back exactly to EM-GS. This is a designed behaviour and §IV-D shows it is load-bearing.

Why not estimate the angles directly? Subspace methods such as root-MUSIC and ESPRIT recover $\{\psi_{\ell,k}\}$ from the signal subspace, but that subspace requires either multiple snapshots or the complex field. With magnitude-only single-snapshot data it is identifiable only *after* phase retrieval has succeeded, at which point the channel is already estimated. We therefore regularise toward the exponential-sum manifold rather than parameterising it.

Related work. Structured low-rank Hankel recovery for sums of exponentials recovers such a matrix from *partial linear* observations with convex relaxations and recovery guarantees: EMaC [6], ALOHA [7] and LORAKS [8]. Gridless line-spectrum and atomic-norm methods [9] likewise assume linear measurements and a separation condition. Both families differ from the present setting in the same way: our measurement is a *modulus*, so the map is not linear, no convex relaxation applies, and the projection is used as a regulariser inside a non-convex iteration rather than as the estimator. We claim no new projection algorithm; Cadzow's [10] is used as given.

Statistics. The primary statistic is the paired per-trial median within an SNR bin, with bootstrap 95% intervals. Where a curve is compared to a bound we use the ratio of summed errors, which is what a bound on the mean-squared error governs; both are reported for the headline cell.

III. Performance Benchmark

A. Rank-one Fisher information

Conditional on $\mathbf{G}, \mathbf{S}, \mathbf{B}$ the noiseless field at (n, p) is a constant λ, so $z = |\lambda + w|$ is Rice distributed and its density depends on λ *only through* $a = |\lambda|$. With $\frac{d}{dx}\log I_0(x) = R(x)$ and $\kappa = 2za/\sigma^2$, the score is $\partial_a \log p = \frac{2}{\sigma^2}[zR(\kappa) - a]$; writing $\beta = \sigma^{-4}(\mathbb{E}[z^2R^2(\kappa)] - a^2)$, one measurement carries scalar information 4β, so in real coordinates $\boldsymbol{\eta} = [\Re\mathrm{vec}\,\mathbf{G}; \Im\mathrm{vec}\,\mathbf{G}]$,

$$\mathbf{J}_n = \sum_{p=1}^{P} 4\beta(a_{n,p}, \sigma^2)\, \mathbf{u}_{n,p}\mathbf{u}_{n,p}^T. \tag{4}$$

Each magnitude measurement contributes rank-one information, not rank two. Crediting both quadratures overstates it and gives a bound too low by 0.17–4.64 dB across our 36 points. An identifiability check settles the form: at $K = 3$, $P = 3$ the per-row problem is unidentifiable and (4) is correctly singular, whereas the rank-two construction claims full rank.

Because row n of $\mathbf{GS}$ enters only measurement row n, $\mathbf{J}$ is *block diagonal across receive elements*. **This is Step 1 of the argument**, and it is why an unstructured estimator — and any estimator attaining the unconstrained bound — is flat in N under a normalised metric.

B. Constrained bound on the geometric manifold

The unconstrained bound governs estimators treating $\mathbf{G}$ as $2NK$ free parameters. HS-GS exploits (2), whose parameters $\boldsymbol{\phi} \in \mathbb{R}^{3\sum_k L_k}$ number about 45, *independent of* N. The textbook form $\mathbf{D}(\mathbf{D}^T\mathbf{J}\mathbf{D})^{-1}\mathbf{D}^T$ with $\mathbf{D} = \partial\boldsymbol{\eta}/\partial\boldsymbol{\phi}$ is unusable, $\mathbf{D}$ being rank deficient: at $N = 8$ its mean numerical rank is 40.4 against ≈ 44.7 parameters. We use the Gorman–Hero / Stoica–Ng tangent-space form [13], [14] with $\mathbf{U}$ an orthonormal basis of $\operatorname{range}(\mathbf{D})$:

$$\mathrm{CCRB} = \mathbf{U}(\mathbf{U}^T\mathbf{J}\mathbf{U})^{-1}\mathbf{U}^T, \tag{5}$$

which depends on the tangent *space* and is invariant to over-parameterisation. Both bounds average over the same realisations as the estimator curves.

TABLE I

THE FOUR CONFIGURATIONS. ALL USE THE *same* ESTIMATOR AND THE SAME HELD-OUT ORDER RULE ($\nu = 0.3$, CANDIDATES $1..r_{\max}$, NO ORACLE) AND ALL USE $n_{\mathrm{cz}} = 4$ CADZOW SWEEPS; THEY DIFFER IN OPERATING POINT. GEN.: TB = TRACK B ULA GENERATOR, TD = TRACK D GENERATOR. "$\mathcal{U}$" MEANS SNR DRAWN PER TRIAL AND BINNED POST HOC. †IN CONFIGURATION C P IS ITSELF THE SWEPT VARIABLE IN THE K-INVARIANCE CELLS (13–27) AND THE PILOT-COUNT CELLS (10–35); 20 EVERYWHERE ELSE.

	T	sel.	P	RSR	Gen.	trials	SNR (dB)	used in
A	100	25	10, 30	12	TB	300–400	$\{-5..20\}$	§IV-A, §IV-B, §IV-F
B	50	20	30	12	TB	400	$\{-5..20\}$	§IV-F (timing)
C	100	25	20†	10	TD	60–1000	$\mathcal{U}[-10, 20]$	§IV-D–§IV-F
D	50	20	30	12	TB	300	5	§IV-C, §IV-D

Bias caveat. Both bounds constrain the covariance of *unbiased* estimators. All estimators here run to a fixed budget with a shrinking update, and HS-GS additionally truncates rank, so none is unbiased and neither bound applies strictly. The CCRB is the primary reference in Fig. 1 because it is the one matched to the estimator's model class; the unconstrained bound is plotted as a secondary line and governs EM-GS only.

IV. NUMERICAL RESULTS

Both estimators are always handed identical realisations, and $K = 3$ except where K is the swept variable. Everything else varies by configuration, so Table I states the four in use and every number below carries its label. Two of them (A and B, and separately C and D) share an operator and differ only in operating point; A and B differ only in T and the selection budget, at one cell measured both ways a difference of 0.005 dB.

A. Main result against the bounds

Configuration A, $N = 32$, $P = 30$. HS-GS improves on EM-GS by $+2.46$ to $+3.05$ dB across the sweep (six-point mean $+2.68$ dB; ratio-of-sums $+2.33$ to $+2.96$ dB), and the projection is active in 100% of trials at this cell (Fig. 1).

The benchmark makes that number interpretable. EM-GS tracks the unconstrained bound to within 0.143 dB for SNR ≥ 5 dB, so the bound is informative rather than vacuous. The geometric CCRB lies 7.05–7.11 dB *below* it at this configuration: that gap is what the structural model is worth in principle, and HS-GS recovers part of it, remaining above the CCRB at all six points by 4.39–4.93 dB.

B. Aperture scaling, and which statistic reverses

Step 4 predicts the gain turns negative at small aperture. We swept $N \in \{8, 16, 32\}$ over 12 operating points each (2 pilot counts $\times$ 6 SNR values) at **configuration A**, so that this panel and Fig. 1(a) share one operator. **The two pooling rules disagree at** $N = 8$ **and agree everywhere else**, so both are reported:

N	ratio of sums	paired median	EM-GS NMSE
8	$\mathbf{-0.254}$	$\mathbf{+0.225}$	-8.705
16	$+0.966$	$+1.081$	-8.672
32	$+2.934$	$+3.226$	-8.700

EM-GS varies by at most 0.163 **dB across the same fourfold change in aperture** — 0.163 dB at $P = 10$ and 0.097 dB at $P = 30$, against a structured gain that moves by over 3 dB. (Pooling the pilot counts first gives 0.033 dB, but they move in opposite directions, so we quote the larger figure.) This is exactly what §III requires of an estimator whose information is block diagonal in N.

The predicted reversal is a ratio-of-sums phenomenon. At $N = 8$ the projection is inactive in 41.8% of trials on average, rising from 12% at -5 dB to 59% ($P = 10$) and 79% ($P = 30$) at $+20$ dB as the selector declines a constraint it cannot afford. Where it is active it truncates genuine components, and the ratio of summed errors turns over accordingly, running from $+1.272$ dB at -5 dB SNR to -2.613 dB at $+20$ dB ($P = 10$), because truncation bias does not shrink with noise while the variance term does. *That is the predicted failure, measured.*

The paired median does not reverse, and we do not claim it does. At $N = 8$ it is $+0.000$ dB at nine of the twelve points, positive and significant at the other three, and negative-and-significant at none. The mechanism is the one §IV-D sets out: abstention makes most trials bit-identical to EM-GS and pins the median at zero, while the active minority dominates a sum of squared errors. So Step 4's prediction is confirmed on the statistic a bound governs and is *invisible* to the one we otherwise treat as primary. A reader who took only the median away would conclude the prior is never harmful; §IV-D shows it is not.

P22, registered before this sweep ran, failed. The predicted bands were $[-0.35, +0.05]$, $[+0.60, +0.95]$ and $[+2.65, +3.00]$ dB: $N = 8$ and $N = 32$ landed inside, $N = 16$ missed by 0.016 dB. The EM-GS spread was predicted at ≤ 0.15 dB: 0.033 dB pooled, which is what the prediction named, but 0.163 dB at $P = 10$ alone, which would fail it. A failed prediction licenses no reporting but the measurement.

C. Path count is the wrong coordinate

Configuration D. Holding $N = 32$ and varying only the path count, with L identical across users at SNR $= 5$ dB, the gain decays monotonically from $+7.04$ dB at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$, while EM-GS moves by 0.191 dB in total. So the operative variable is channel complexity relative to capacity, and aperture matters only because it sets the capacity.

D. Effective rank, and what the prior does versus the estimator

Path count is still imperfect: clustered or unequal-strength paths occupy fewer effective dimensions. We index by the Roy–Vetterli effective rank [12] of the lifting, $r_{\mathrm{eff}} = \exp(-\sum_i p_i \ln p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$, normalised by capacity, $\rho = r_{\mathrm{eff}}/r_{\max}$.

Re-indexing the sweep of §IV-C (configuration D) moves the zero crossing from $L/r_{\max} = 0.90$ to $\rho = 0.518$, and the useful region becomes $\rho \lesssim 0.5$ — a statement about any channel rather than one simulator's L values. Across $N \in \{16, 32, 64\}$ the crossing is 0.588, 0.518, 0.544, a spread of 0.070 over a fourfold change in aperture (Fig. 2). **This one**

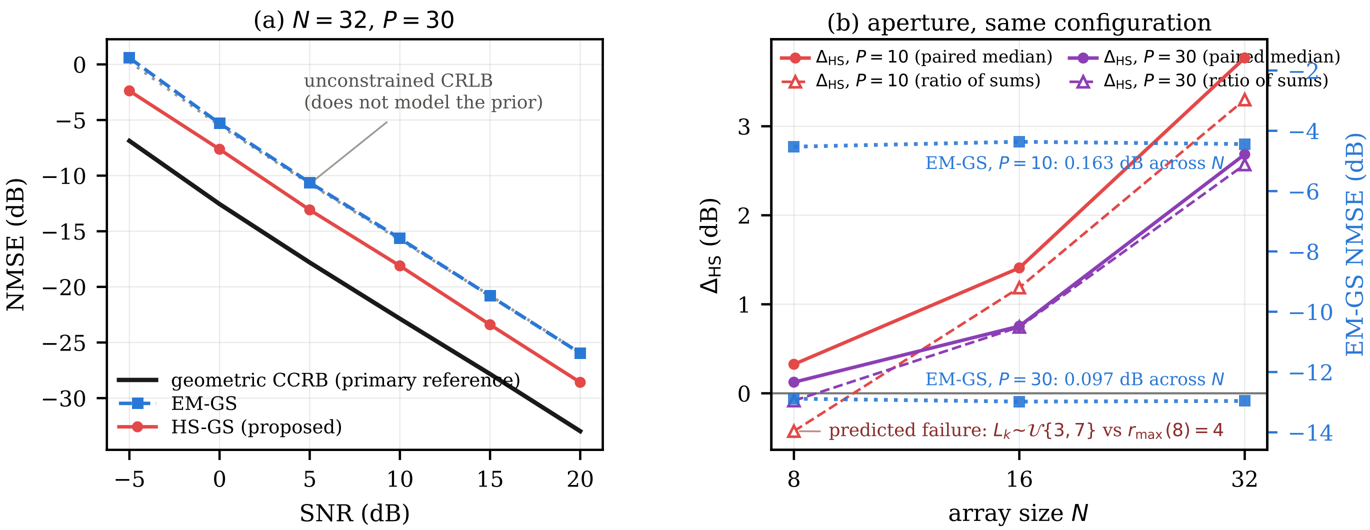

Fig. 1. **Both panels are configuration A** — $n_{cz} = 4$, $T = 100$, selection budget 25, RSR 12 dB, $K = 3$, 400 paired trials per point — so the aperture panel and the benchmark panel share one operator. **(a)** NMSE at $N = 32$, $P = 30$. The geometric CCRB is the primary reference; the unconstrained CRLB is faint because it does not model the prior and governs EM-GS only. Neither bound governs a biased estimator. **(b)** Aperture. Δ_{HS} against N at both pilot counts, each point the mean over the six SNR values; filled markers are the paired per-trial median (primary), open markers the ratio of summed errors, and the two disagree at $N = 8$. EM-GS's own NMSE is overlaid on the right-hand axis, whose span is fixed at ± 3 dB so that a flat line reads as flat: the structured curve moves across aperture and the unstructured one does not.

comparison spans two configurations: the $N = 32$ value is configuration D and the other two are configuration C, and only the $N = 32$ crossing is bracketed by measurements on both sides — the other two are extrapolated from the positive side, for the abstention reason given below.

The crossing belongs to the prior, not to the estimator, and the two must be stated separately. Pushing past the crossing under configuration C, the paired per-trial median never goes negative: at $\rho = 0.608$ and 0.735 ($N = 16$) it is $+0.078$ and $+0.000$ dB, and at $\rho = 0.596$ and 0.669 ($N = 64$), $+0.044$ and $+0.018$ dB. The reason is abstention: at $\rho = 0.735$ the selector returns $\hat{r} \geq r_{max}$ in 38.8% of trials, and where it does the estimator is exactly EM-GS. Forcing the projection active at that same cell gives -0.075 dB, CI $[-0.186, -0.026]$. So the prior does become harmful past the crossing, and the abstention rule is what prevents the *typical* trial from following it down.

This does not contradict the negatives reported above, and the difference is the statistic. §IV-B and §IV-C report ratio-of-sums values that do go negative. Abstention makes the majority of trials bit-identical to EM-GS, which pins the *median* at exactly zero; it does nothing about the minority in which the projection is active and truncates genuine components, and those trials dominate a sum of squared errors. §IV-B is the same effect at $N = 8$. **So $\Delta_{HS} \geq 0$ is a property of the selector under a median statistic only. It is neither evidence that the prior is never harmful nor a guarantee about mean-square error**, and where the two statistics disagree we report both.

The gain magnitude does not collapse onto the same axis. Over the window both arrays cover (configuration C), $N = 16$ lies above $N = 64$ by a mean of 0.493 dB and a maximum of 0.901 dB, one-signed throughout. *We have no account of it.* The one candidate we tested — that the selector's rank grid is coarser at small capacity — is refuted: varying the pencil at fixed $N = 32$ and fixed ρ gives a span of 0.390 dB over admissible-rank counts 8 to 16 with *no monotone trend*, and the dependence runs mildly backwards. A pair with opposite matrix shapes but nearly equal rank counts (24×9 and 8×25) gives $+3.199$ and $+3.121$ dB, so the effect tracks the rank count and not the shape.

ρ is a design-time coordinate, not a receiver-observable one, since r_{eff} is computed on the noiseless channel. The selector's abstention rate is observable and is monotone in ρ at fixed aperture, but it is confounded by capacity: at $N = 64$ a cell at $\rho = 0.501$ abstains in 0.0% of trials while at $N = 16$ a cell at $\rho = 0.371$ abstains in 4.0%, so no single threshold separates the useful region. An observable proxy therefore exists within an aperture but has not been established across apertures.

E. Cross-generator prediction

A relation fitted and evaluated on the same data proves little. We froze the ρ-to-gain relation obtained above, committed numerical predictions to version control, and only then ran two channel generators that played no part in constructing it. The rule is piecewise-linear interpolation on a fixed eight-point table — no regression, no spline — so it re-executes exactly. Both tests run configuration C's operator and operating point with the generator replaced.

Clustered Saleh–Valenzuela. At $\rho = 0.331$ the relation predicted $+1.30$ dB; the measurement over 276 trials gave $+1.173$ dB, CI $[+1.017, +1.380]$ — an error of 0.13 dB.

A second, independently specified configuration. At $\rho = 0.400$ the prediction was $+0.648$ dB with interval $[+0.367, +1.015]$; the measurement over 266 trials gave $+0.963$ dB, CI $[+0.808, +1.086]$ — an error of 0.315 dB, inside the stated interval but on its upper edge.

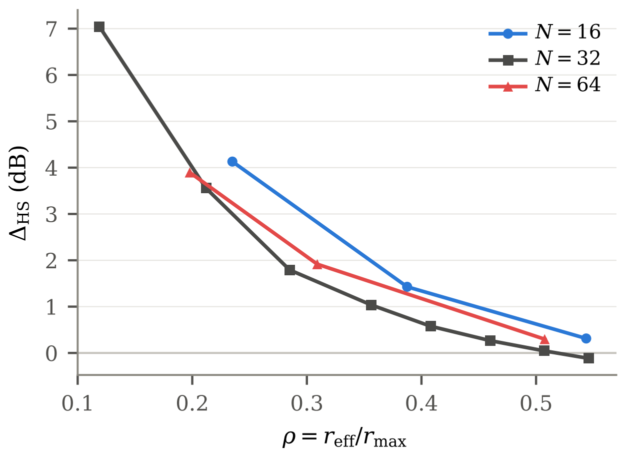

Fig. 2. Δ_{HS} against $\rho = r_{\mathrm{eff}}/r_{\max}$ for $N \in \{16, 32, 64\}$. The boundary *location* is stable across a fourfold change in aperture; the gain *magnitude* is not, and that residual is unexplained.

Scope. These are *cross-generator* tests, not out-of-model tests: both generators still produce channels of the form (2), and only the distribution over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$ changes. They show ρ transfers across path distributions, not that it survives a violation of the ULA model.

F. Users, cost, and where the projection belongs

K-invariance (configuration C). At fixed pilot adequacy $P/2K = 3.33$ the gain varies by 0.094 dB across $K \in \{2, 3, 4\}$ for SNR ≥ 5 dB. Pooled over all SNR the spread is 0.294 dB against a pre-registered falsifier of 0.3 dB — missed by 0.006 dB, and reported rather than dropped.

Cost. Order selection is 57.7–85.5% of runtime. A coarse-to-fine stride-3 search (configuration C) cuts the selection stage by $1.68\times$ ($16.0 \to 9.6$ evaluations) and returns the same $\hat{r}$ in 93.3% of trials, but **missed its pre-registered gate**: the criterion was ≤ 0.05 dB and the aggregate cost is 0.083 dB. Its paired median degradation of 0.000 dB understates that, for the same reason abstention does above, so we report it as an option that missed its gate and not as a validated substitute.

Placement is SNR-dependent, and the effect is small (configuration A). Comparing interleaved projection against a single post-hoc Cadzow projection at the same selected order, interleaving is *worse* at 0 dB (-0.164 dB, CI $[-0.317, -0.066]$), indistinguishable from $+5$ to $+15$ dB, and better only at $+20$ dB ($+0.168$ dB, CI $[+0.102, +0.246]$); the six-point mean is $+0.003$ dB. Interleaving is therefore not required in this classical iteration, and we do not claim it is — a single projection recovers nearly all of the benefit at a fraction of the cost.

What this leaves as the contribution. If placement is a wash, then what the estimator contributes is not *where* the projection is applied but *whether it is applied at all, and at what order* — decided from held-out pilots with no oracle. §IV-D is the evidence: the selector is what keeps $\Delta_{\mathrm{HS}} \geq 0$, and forcing the projection past the crossing costs -0.075 dB, CI $[-0.186, -0.026]$. The order selector is the claim, not a mechanism that gets explained on the way to one.

V. Limitations and Conclusion

A magnitude-only receiver's information is rank one per measurement and block diagonal across elements, so aperture is usable only through cross-element structure. The ULA Hankel prior supplies it, with capacity $\lceil N/2 \rceil$; the prior therefore helps only below a fraction of that capacity, and the predicted sign reversal at small aperture is observed. The governing coordinate is $r_{\mathrm{eff}}/r_{\max}$, whose boundary location is stable across a fourfold change in aperture and which predicted the gain on two unseen channel distributions to 0.13 and 0.315 dB.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover geometric ULA channels of the form (2) at the array sizes, pilot counts and generators tested, and the cross-generator tests change the distribution but not the model family; no claim is made beyond them. The gain magnitude does not collapse onto the proposed axis and its residual of 0.493 dB is unexplained. ρ is not receiver-observable. Two of the three boundary crossings cannot be bracketed from the negative side by the proposed estimator, because abstention prevents it; $\rho \approx 0.75$ is in any case unreachable at $N = 64$, where the path generator fails above $L = 48$ ($\rho \approx 0.67$). We establish no fundamental bound: the CRLB and CCRB govern unbiased estimators and every estimator here is biased.

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